>head(mtcars)

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
mazda RX4 wag	21.0	6	160	110	3.90	2.874	17.02	0	1	4	4
Datsun 710	22.8	6	108	93	3.85	2.370	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
Hornet sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	4	255	105	2.76	3.460	20.22	1	0	3	1

Ex:No: 7 Date: 16/04/26	Implementation of Linear Regression Using mtcars Dataset

Problem Description:

The mtcars dataset contains specifications of different car models, including fuel efficiency and engine characteristics. To understand how engine and vehicle attributes influence fuel efficiency, linear regression can be used to model the relationship between **miles per gallon (mpg)** and predictors such as **engine displacement (disp), horsepower (hp), and weight (wt)**.

Objective:

To implement a linear regression model in R to analyze the relationship between fuel efficiency and vehicle attributes, interpret the regression coefficients and model summary, and visualize the regression results for better understanding of the model.

Procedure:

- **Step 1: Load Required Libraries**

  We load the necessary libraries for implementing linear regression.

```
# Load required libraries
install.packages ("ggplot2")        library (ggplot 2)   // data visulizat
install.packages ("caTools")        library (caTools)    // data splitting
install.packages ("plotly")         library (plotly)     // interactive
```

- **Step 2: Load and Prepare the Dataset**

  We will use the built-in mtcars dataset and explore it.

```
# Load the dataset
data(mtcars)

# Display first few rows
head (mtcars)

summary (mtcars)
```

Dataset Explanation:

mpg (Miles per gallon) → Dependent Variable (Y)

hp (Horsepower) → Independent Variable (X)

We will predict mpg based on hp using linear regression.

>summary(mtcars)

	mpg	cyl	disp	hp	drat	wt
min :	10.70	min: 4.00	min :71.1	min : 52.0	min: 2.760	min: 1.513
1st Qu :	15.43	1st Q : 4.00	1st Q : 120.8	1st Q : 96.5	1st Q: 3.080	1st Q: 2.581
median :	19.20	median: 6.000	median: 196.3	median: 123.0	median: 3.695	median: 3.325
mean :	20.09	mean : 6.188	mean : 230.17	mean: 146.7	mean: 3.597	mean: 3.217
3rd Qu:	22.80	3rd Qu: 8.000	3rd Qu: 326.0	3rd Qu: 180.0	3rd Qu: 3.920	3rd Qu: 3.610
max :	33.90	max: 8.000	max : 472.0	max : 335.0	max: 4.930	max: 5.424

>summary(model.simple)

call :

lm(formula = mpg ~ hp, data = train.data)

Residuals:

min	1Q	median	3Q	max
-5.2038	-2.078	-0.4764	1.8764	7.738

Coefficients:

	Estimate	std. Error	t value	Pr(>|t|)	
(Intercept)	31.67218	2.03579	15.558	1.23e-12	***
hp	-0.07670	0.01313	-6.068	6.27e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error : 3.603 on 20 degrees of freedom

multiple R-squared: 0.648 Adjusted R-squared : 0.6307

F-statistic : 36.82 p-value : 6.237e-06

- **Step 3: Split the Dataset into Training and Testing Sets**
  We split the dataset into 70% training and 30% testing.

  # Split data into training (70%) and testing (30%)

```
Set.seed(123)
split <- sample.split (mtcars$mpg, splitRation = 0.7)
train_data <- subset (mtcars, split = =TRUE)
test_data <-subset (mtcars, split = =FALSE)
```

- **Step 4: Simple Linear Regression (mpg vs hp)**
  We build the linear regression model using lm() function.

  # Train linear regression model

```
modelsimple <- lm(mpg~hp, data = train_data)
```

  # Print model summary

```
summary (model simple)
```

- **Step 5: Make Predictions on Test Data**
  We use the trained model to make predictions on the test dataset.

  # Make predictions on test data

```
pred <- predict (model_simple, newdata = test_data)
```

  # Display predicted values

```
pred
```

> pred

	Hornet 4 Drive	Hornet Sportabout	Merc 240D	Merc 280C
28	22.905341	17.729928	26.730867	21.86760

Maserati Bora	Volvo 142E
4.673177	22.985037

> MSE: 20.75469

> R-squared : 0.5672003

> summary (model_mul)

Call:

lm (formula = mpg ~ wt + hp + disp , data = train_data)

Residuals :

min	1Q	median	3Q	Max
-3.6776	-1.6313	-0.1022	0.5424	5.8629

Coeff :

	Estimate	std. Error	t value	Pr(>|t|)
(Intercept)	31.67218	2.03579	15.558	1.23e-12 ***
wt	-3.263180	1.311567	-2.486	0.0229 *
hp	-0.033409	0.019148	-1.745	0.0981 .
disp	-0.003653	0.014111	-0.259	0.7986

Signif . codes: 0 '***' 0.001 '**' 0.01 '*' , 0.05 '.' 0.1 ' ' 1

F-statistic : 27.65 on 3 and 18 DF , p-value : 5.882e-07 .

- **Step 6: Evaluate Model Performance**

  We compute Mean Squared Error (MSE) and R-squared for evaluation.

  # Compute Mean Squared Error (MSE)

```
mse <- mean((test data $mpg- predictions)^2)

cat ("MSE :", mse, "\n")
```

  # Compute R-squared value

```
R-squared <- cor(test-data$mpg, predictions)^2)

cat ("R-squared :", R-squared, "\n")
```

- **Step 7: Visualize Classification Results**

  We plot the regression line on a scatter plot.

  # Scatter plot with regression line – hp vs mpg

```
ggplot (train_data, aes (x = hp, y= mpg)) +
    geom_point (color = "blue", size= 3)+
    geom_smooth (method = "lm", col = "red")+
    ggtitle (" Linear Regression : HP VS MPG")
```

- **Step 8: Multiple Linear Regression** - mpg – wt, hp, disp

```
model_mul <- lm (mpg ~wt + hp+ disp, data = train_data)
```

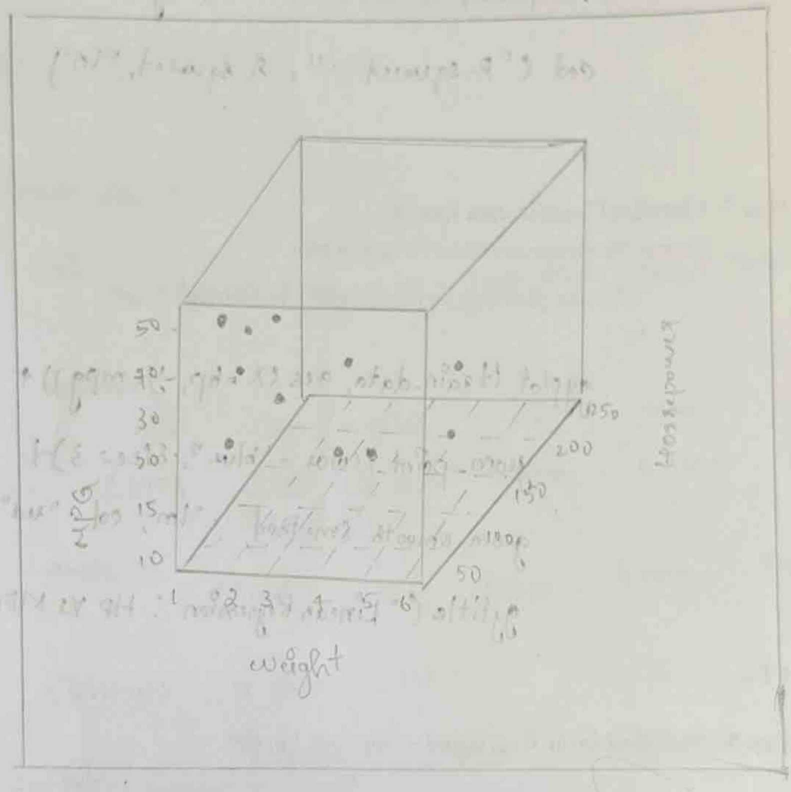
  # Model summary

```
summary (model_mul)
```

>head(pred)

Mazda RX4   Mazda RX4 Wag   Datson 710   Valiant   Dust 360

23.39958   22.56747   25.13647   20.58808   15.05864   22.2418



- **Step 9: Multiple Linear Regression - Prediction**

  # Predicted mpg values

```
pred <- predict(model_multiple)

head (pred)
```

- **Step 10: 3D Visualization (mpg vs wt and hp)**

```
library (scatterplot 3d)

Scatterplot 3d (mtcars $wt, mtcars$hp, mtcars $mpg,
        xlab = "weight",
        ylab = "Horsepower"
        zlab = "MPG"
        main = "3D plot of MPG Vs weight and Horsepower")
```

Result:

Thus, the regression analysis demonstrates that vehicle weight, horsepower, and engine displacement have a significant negative effect on fuel efficiency. The multiple linear regression model provides better explanatory power than simple regression and successfully predicts miles per gallon based on vehicle characteristics.